

Thomas Hart

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Education

University of Waterloo · BSc in Computer Engineering · 3.71 GPA Sept 2021 – Apr 2026

- **Relevant Courses:** Algorithms and Data Structures (C++), Systems Programming and Concurrency (C), Distributed Computing, Real-Time Operating Systems (C, ARM), Computer Architecture (Verilog)

Skills

Languages: C++, C, Python, Rust, Java, TypeScript, JavaScript, PostgreSQL, MySQL, ARM Assembly, Verilog

Frameworks/Libraries: Pandas, SQLAlchemy, Langchain, React.js, Django, Android SDK, AOSP

Tools: Linux, Git, CMake, Make, Tableau, Metabase, DBT

Experience

Software Engineer · Radox · Antrim, Northern Ireland Sept 2025 – Dec 2025

- Rearchitected a **Python** biochip viewing application using **distributed computing** with **Dask** to eliminate concurrent user stalling and improve scalability for multiple simultaneous users.
- Implemented a **Prefect** server with reverse proxy through **NGINX** to distribute and schedule critical services, streamlining workflow automation and improving system reliability.
- Optimized backend database queries and indexing strategies to reduce load times by **95%+** for critical internal applications using **SQL Server** and **SQLAlchemy**.

Data Engineer · RxFood · Toronto, ON Jan 2025 – Apr 2025

- Built a reporting engine using **Python** and **PostgreSQL** to automatically generate patient outcome reports for clinicians, **saving 2–3 hours** of manual work per report and improving long-term scalability.
- Added **interactive visualizations** to the clinician portal using **React.js** and **TypeScript**, enabling actionable insights into patient health trends for **600+** clinicians.

Software Engineer – Enterprise Co-op · AI Docs · Vancouver, BC Jan 2024 – June 2024

- Built a codebase documentation tool using **Python**, **Langchain**, and abstract syntax tree parsing to automatically generate structured documentation; won the **Overbeeke Family Entrepreneurship Excellence Award (\$5,000)** from a university pitch competition out of **30+** students.

Undergraduate Research Assistant · University of Waterloo · Waterloo, ON May 2023 – Aug 2023

- Performed **formal verification** of **Rust** code using the **SeaHorn** verification framework, achieving a **50%** reduction in verification time by implementing alternatives to standard library data structures.

Software Engineer · Peraso Technologies · Toronto, ON Sept 2022 – Dec 2022

- Developed a custom **XML** parsing and generation tool in **C++** to streamline **EEPROM** programming, achieving **95%** faster input speed compared to manual entry.
- Optimized **CLI firmware** for **5G radio devices** and refactored redundant command outputs, resulting in **20%** faster runtime and a **10%** reduction in memory usage.

Autonomous Vehicle Android Developer · Ford Motor Company · Remote Jan 2022 – Apr 2022

- Migrated legacy **AOSP** code to the latest **Android** version and a newer build system for an in-vehicle infotainment system, enabling long-term compatibility and reduced build times.

Projects

Pre-Trade Risk Gateway System (Verilog, FPGA, Linux, C)

- Developed a pre-trade risk gateway on an **AMD KV260 FPGA** to perform real-time risk checking for high-frequency trading on the NASDAQ, achieving end-to-end latency of **5.5 microseconds** with guidance from Jane Street engineers.
- Wrote **Verilog** RTL, built a custom **Vivado** block design and implemented a **Linux** kernel driver for DMA-based communication between the processing system and FPGA fabric over Ethernet.

Finance Translator (Python, Django, HTML, CSS)

- Developed a web application using **Django** to simplify complex financial text using **NLP** through the **Cohere API**.
- Implemented a **trie-based search algorithm** to identify keywords in the text and provide relevant links to definitions and explanations.

